

COMICCRAFT – TEST PLAN

1. Objective

The objective of the Test Plan is to define the testing approach used to verify that the ComicCraft application works correctly.

Testing is performed to identify errors, verify functionality, and ensure that the application produces the expected comic output.

2. Scope of Testing

The testing covers the major components of the ComicCraft application:

- User Interface
- Story Prompt Input
- FastAPI Backend
- Gemini AI Integration
- Story Generation
- Image Generation
- Comic Layout
- PDF Generation
- API Communication
- Error Handling

3. Testing Objectives

The main objectives are:

- Verify that the application starts correctly.
- Verify that users can enter a story prompt.
- Verify that valid prompts are processed.
- Verify that invalid inputs are handled.
- Verify that Gemini AI generates story content.
- Verify that comic panels are created correctly.
- Verify that images are generated or handled properly.
- Verify that the comic layout is created.
- Verify that the PDF is generated correctly.
- Verify that the final output can be accessed by the user.

4. Testing Types

4.1 Functional Testing

Functional testing verifies whether each feature performs according to its requirements.

Examples:

- Prompt submission
- Story generation
- Panel generation
- Image generation

- PDF generation

4.2 Integration Testing

Integration testing verifies communication between different modules.

Examples:

- Frontend and backend
- Backend and Gemini AI
- Story generation and image generation
- Comic layout and PDF generation

4.3 Usability Testing

Usability testing checks whether the application is simple and easy to use.

The following are checked:

- Clear interface
- Easy prompt entry
- Understandable buttons
- Readable comic output
- Simple PDF download process

4.4 Error Testing

Error testing verifies that the application handles invalid inputs and unexpected problems correctly.

Examples:

- Empty prompt
- Invalid prompt

- Missing image
- AI generation failure
- PDF generation failure

5. Test Environment

The application can be tested using:

- Computer or laptop
- Windows/Linux/macOS
- Python environment
- FastAPI server
- Modern web browser
- Internet connection
- Configured Gemini API access

6. Test Data

The following types of test data can be used:

Valid Input

Example:

"Create a short comic about a student discovering a mysterious book."

Empty Input

No story prompt is entered.

Short Input

"Create a superhero comic."

Longer Input

A detailed story idea containing multiple characters, locations, and events.

7. Expected Testing Result

The application should:

- Accept valid story prompts.
- Reject empty or invalid prompts.
- Generate structured comic content.
- Create appropriate panel information.
- Process image generation.
- Create the comic layout.
- Generate the PDF successfully.
- Display suitable error messages when problems occur.

8. Conclusion

The Test Plan provides a structured approach for testing the ComicCraft application.

Testing helps ensure that the individual modules and their integration work correctly and that the final application provides the expected comic generation workflow.